University Important Questions

1. **Enumerate the layers of the scalp and describe their blood supply, nerve supply, and lymphatic drainage.**
2. **Write a note on the trigeminal nerve.**
3. **Describe the maxillary nerve under the following headings: origin, course, branches, distribution, and applied anatomy.**
4. **Describe the course, branches, and distribution of the mandibular division of the trigeminal nerve.**
5. **Describe the origin, course, relations, and branches of the facial artery.**
6. **Provide an account of the course, intracranial and extracranial path, and applied anatomy of the facial nerve.**
7. **Give a short account of the parts, relations, and nerve supply of the lacrimal gland.**
8. **Describe the levator palpebrae superioris muscle**.
9. **Some important terms**

The most important questions are:

Q1, Q2, Q3, Q4, Q6

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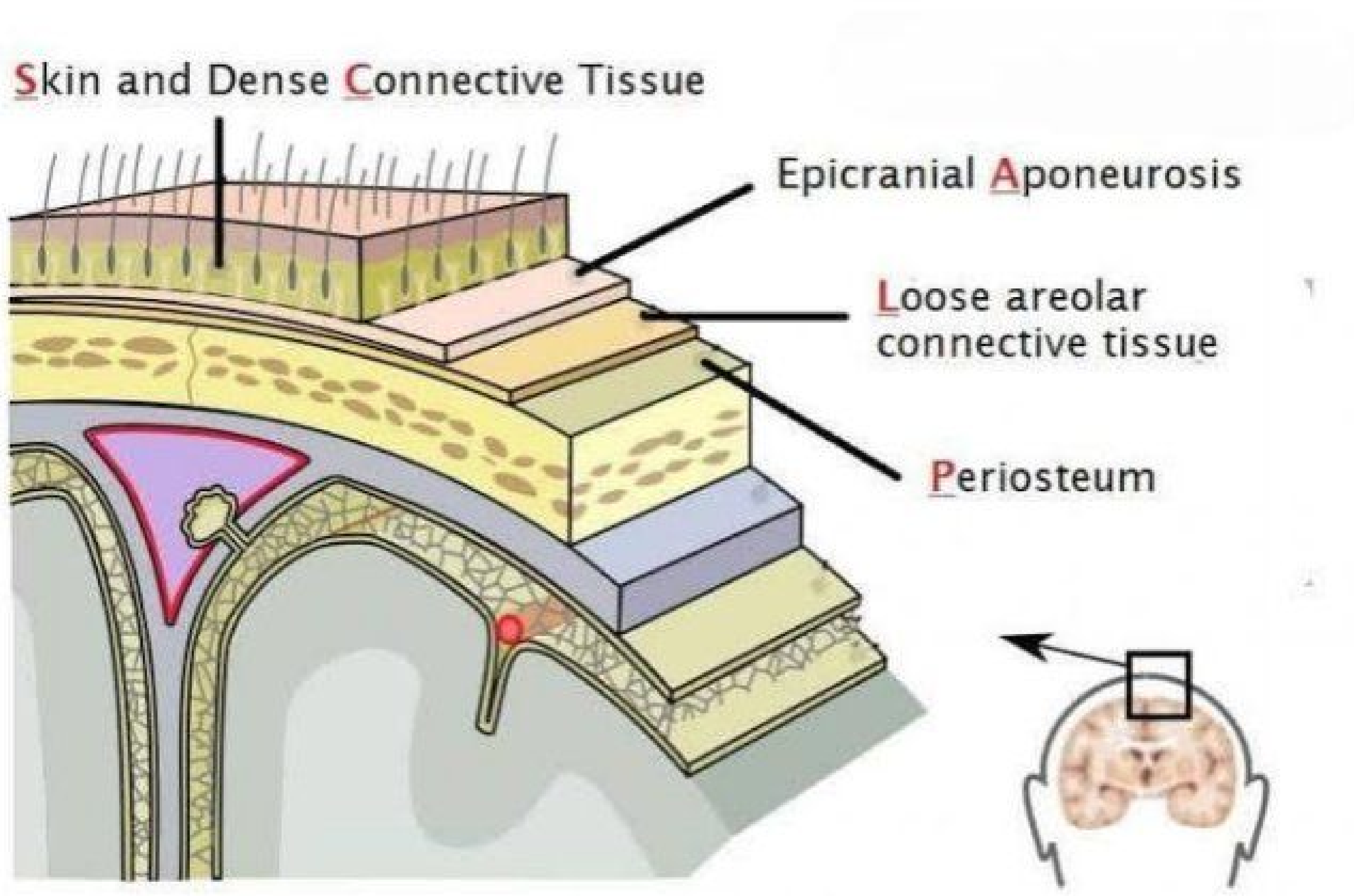
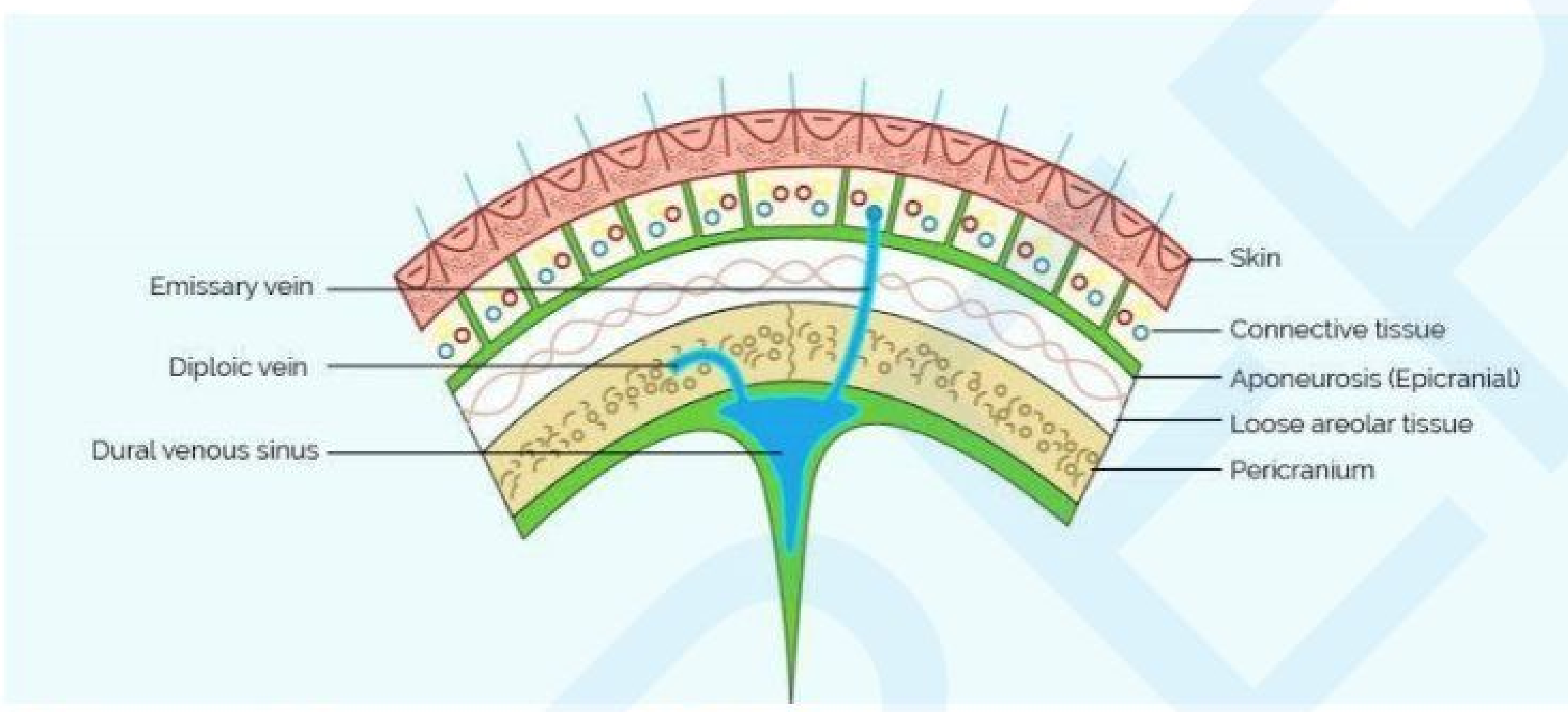
Q1. Enumerate the layers of scalp. Give their, blood supply, Nerve supply and lymphatic drainage

Scalp





5 Layers (Mnemonic: SCALP- **S**kin, **S**uperficial fascia, **D**eep fascia, **L**oose areolar tissue, **P**ericranium)

- **Skin** - Thick, hairy outermost layer, contains **sebaceous glands** & **rich blood supply**, adherent to **epicranial aponeurosis** via **superficial fascia**
- **Superficial fascia** - Fibrous & dense in center, binds **skin to aponeurosis**, allows **passage of vessels & nerves**
- **Deep fascia (Epicranial aponeurosis / Galea aponeurotica)** - Movable on **pericranium**, receives **occipitofrontalis muscle** (frontal & occipital bellies)
  - Attachments of Epicranial Aponeurosis
    - **Anteriorly** - **Insertion of frontalis**
    - **Posteriorly** - **Insertion of occipitalis**, attached to **external occipital protuberance & highest nuchal lines**

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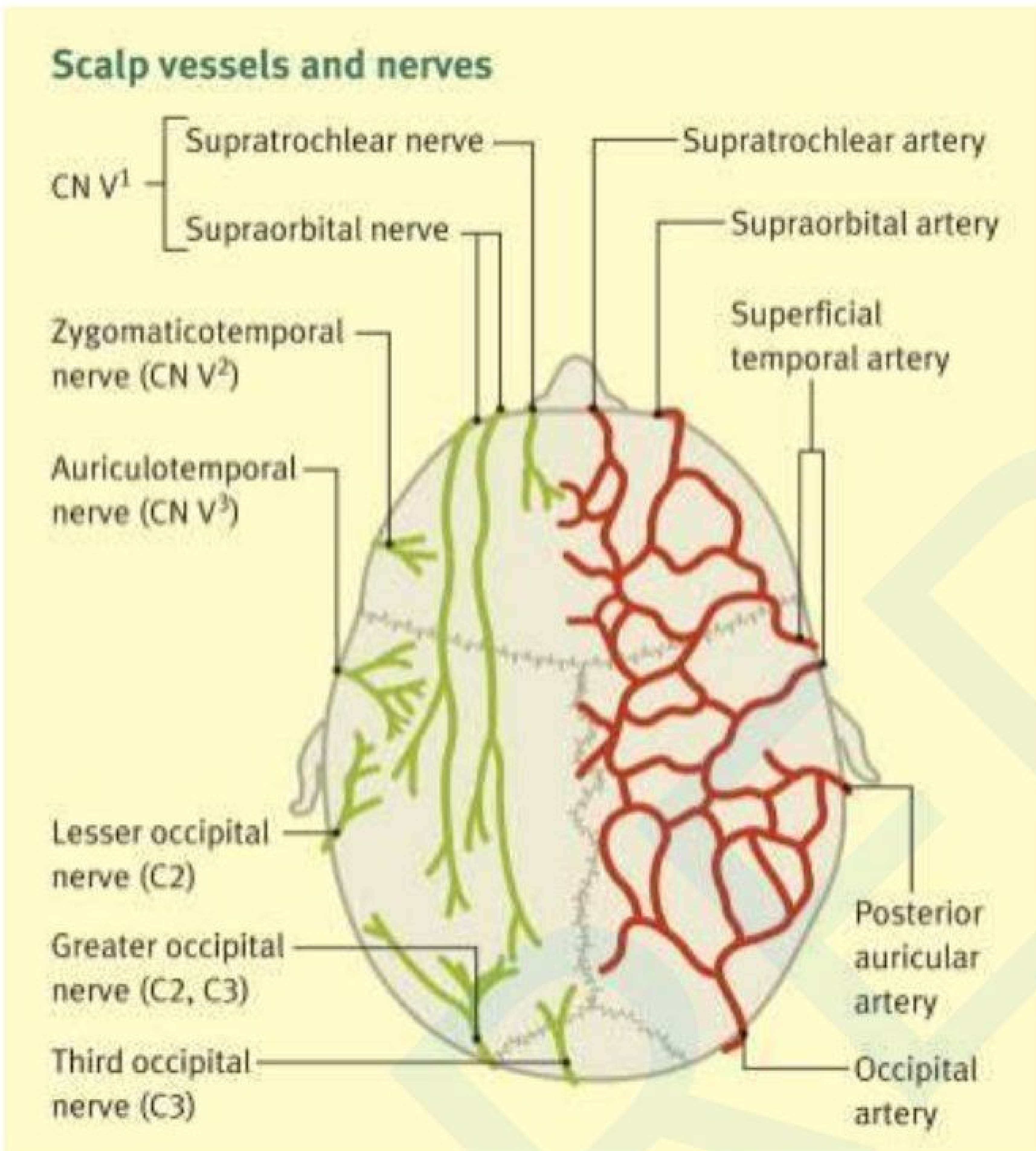
- **Laterally** - Attached to **superior temporal line**, extends down as a thin expansion over **temporal fascia** to **zygomatic arch**

- **Loose areolar tissue** - Extends **anteriorly to eyelids, posteriorly to nuchal lines, laterally to superior temporal lines**, allows free movement of upper 3 layers
- **Pericranium** - Loosely attached to **bones**, but **firmly adherent at sutures** (via **sutural ligaments**), forms outer layer of skull bones

Arterial Supply of Scalp



- **In front of auricle** (supplied from internal & external carotid arteries)
  - **Supratrochlear artery** (from **ophthalmic artery → internal carotid**)
  - **Supraorbital artery** (from **ophthalmic artery → internal carotid**)
  - **Superficial temporal artery** (from **external carotid**)
- **Behind auricle** (supplied from external carotid artery)
  - **Posterior auricular artery**
  - **Occipital artery**

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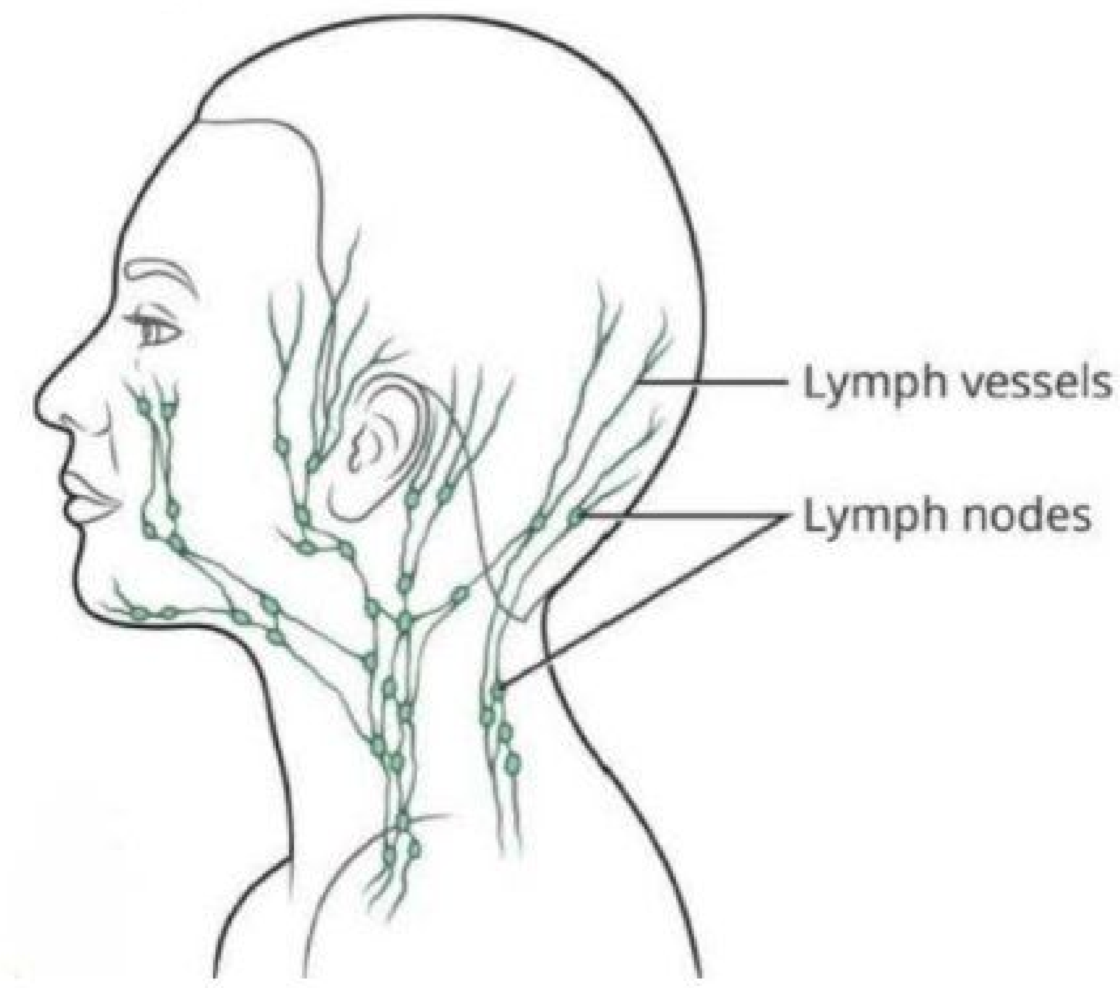
Lymphatic Drainage



- **Preauricular / Parotid LN** → Drain **anterior** part of **scalp**
- **Posterior auricular / Mastoid / Occipital LN** → Drain **posterior** part of **scalp**

Nerve Supply

- **10 nerves each side** → **5 in front** auricle, **5 behind** auricle
- **4 sensory + 1 motor** in both areas

Front of Auricle

- **Sensory Nerves**
  - **Supratrochlear**
  - **Supraorbital**
  - **Zygomaticotemporal**
  - **Auriculotemporal**
- **Motor Nerve**
  - **Temporal branch of facial nerve**

Behind Auricle

- **Sensory Nerves**
  - **Great auricular (C2, C3) - cervical plexus**
  - **Lesser occipital (C2) - cervical plexus**
  - **Greater occipital (C2) - dorsal ramus**
  - **Third occipital (C3) - dorsal ramus**
- **Motor Nerve**
  - **Posterior auricular branch of facial nerve**

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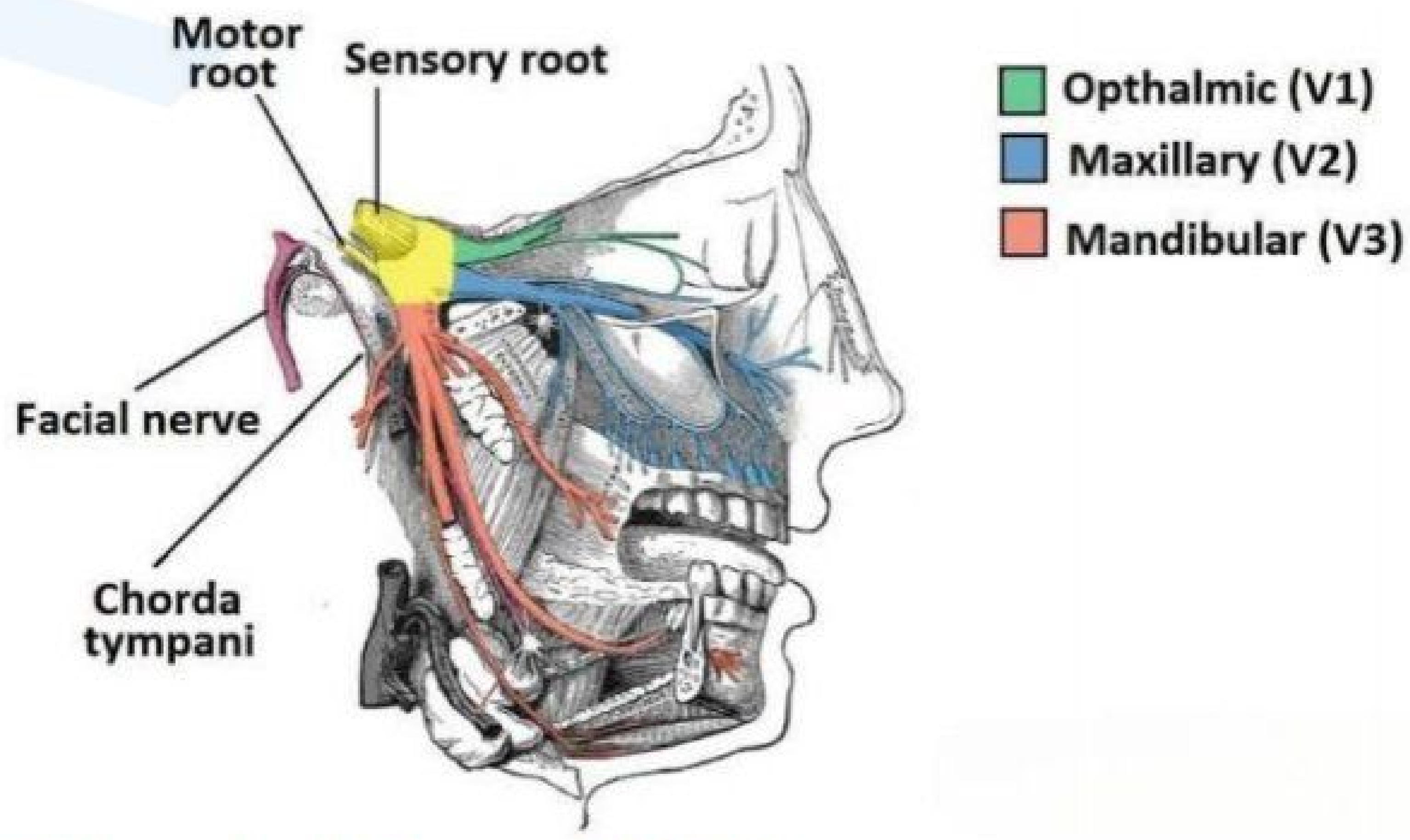
Q2. Write a note on trigeminal nerve.



Trigeminal Nerve (CN V)

- **Origin**: Arises from the **pons** in the brainstem.
- **Components**: It is a **mixed nerve** with both sensory and motor fibers.
- **Divisions**: The nerve divides into three branches:
    - **Ophthalmic (V1)**: Sensory - forehead, eyes, and upper nose.
    - **Maxillary (V2)**: Sensory - maxilla, upper lip, and cheeks.
    - **Mandibular (V3)**: Sensory - lower lip, chin, and jaw; **Motor** - muscles of mastication (e.g., masseter, temporalis).
- **Functions**:
    - **Sensory**: Carries sensation from the **face**, **teeth**, and **sinuses**.
    - **Motor**: Controls muscles for **chewing** (mastication).
- **Pathway**:
    - The **sensory roots** enter the **trigeminal ganglion** and then branch off.
    - The **motor root** joins the mandibular division and exits via the **foramen ovale**.
- **Sensory Ganglion**: Located in the **trigeminal ganglion** in the **middle cranial fossa**.
- **Reflexes**: Involved in the **corneal reflex** and **jaw jerk reflex**.

Q3. Describe the maxillary nerve under the following headings: origin, course, branches ,distribution and applied anatomy.

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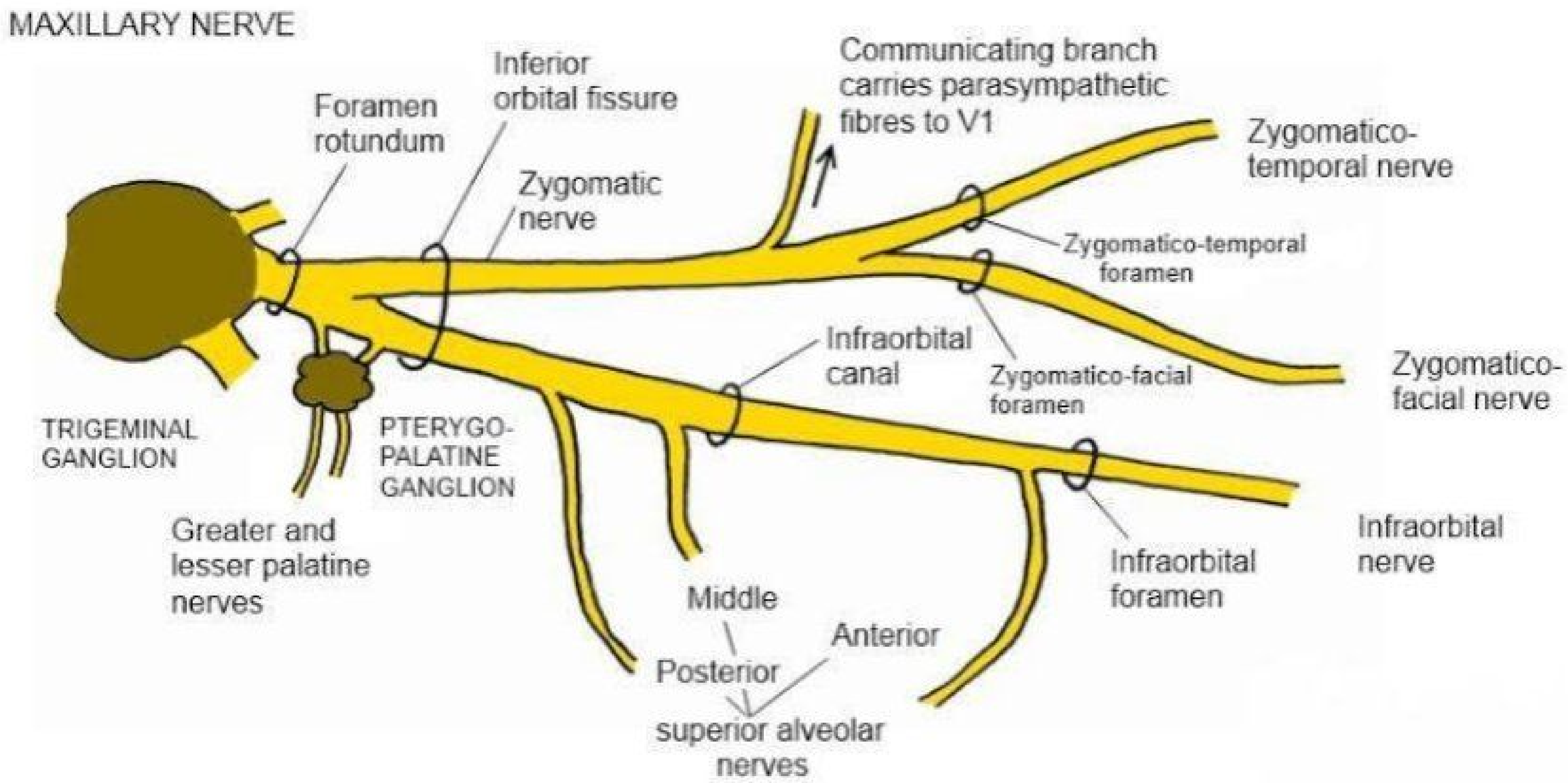
Maxillary Nerve (V2) - Sensory Nerve



- **Origin** - **Anterior border of trigeminal ganglion**
- **Exit from cranial cavity** - **Foramen rotundum** → enters **pterygopalatine fossa**
- **Pathway** - Enters **orbit via inferior orbital fissure**, accompanied by **infraorbital artery**
- **Infraorbital nerve** - Part of maxillary nerve in **floor of orbit**, passes through **infraorbital canal → infraorbital foramen** → divides into **peripheral branches**

Peripheral Branches on Face

- **Inferior palpebral branch** - **Lower eyelid & conjunctiva**
- **Lateral nasal branch** - **Lateral skin of external nose**
- **Superior labial branch** - **Skin & mucosa of upper lip & cheek**

Branches of Maxillary Nerve

Within Cranial Cavity

- **Meningeal branches** - Supply **dura mater of middle cranial fossa**

Within Pterygopalatine Fossa

- **Ganglionic branches** - Join **pterygopalatine ganglion** (carry **sensory & secretomotor fibers**)
- **Zygomatic nerve**
  - **Zygomaticotemporal** - **Temporal region & scalp**, carries **parasympathetic fibers to lacrimal gland**

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    - **Zygomaticofacial** - **Skin of malar region**
- **Posterior superior alveolar nerve** - **Maxillary air sinus, molar teeth, maxillary gum**

Within Floor of Orbit

- **Middle superior alveolar nerve** - Supplies **premolar teeth**, may join **superior dental plexus**
- **Anterior superior alveolar nerve** - **Incisor & canine teeth**, forms **superior dental plexus** with posterior & middle superior alveolar nerves

Ganglionic Branches

- Carry **sensory fibers from pharynx, nose, palate, orbit**
- Receive **secretomotor fibers (via nerve of pterygoid canal) to lacrimal gland**

Zygomatic Nerve

- **Branch of maxillary nerve**, enters **orbit**, divides into **zygomaticotemporal & zygomaticofacial**
- **Zygomaticofacial** - Passes through **zygomatic bone** to supply **malar region**
- **Zygomaticotemporal** - **Temporal region & scalp**, carries **parasympathetic fibers to lacrimal gland**

Superior Alveolar Nerves

- **Posterior superior alveolar nerve** - **Maxillary air sinus, molar teeth, maxillary gum**
- **Middle superior alveolar nerve** - **Premolars, superior dental plexus**
- **Anterior superior alveolar nerve** - **Incisors, canines, nasal wall**

Applied Anatomy

- **Trigeminal neuralgia** - Severe pain along **trigeminal nerve, including maxillary nerve**
- **Cavernous sinus thrombosis** - Blood clot may **compress maxillary nerve** causing symptoms

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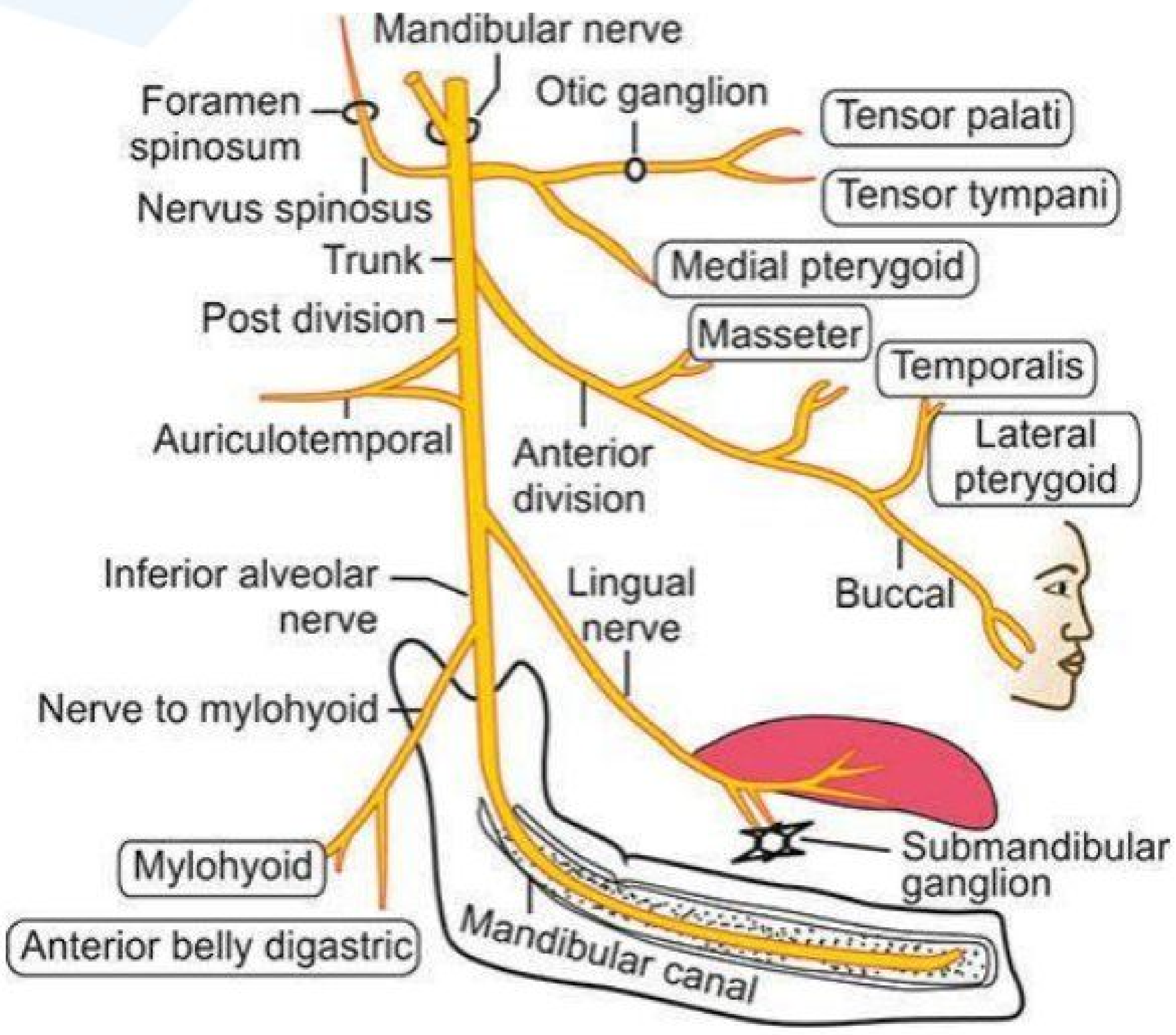
Q4. Describe the course, branches and distribution of mandibular division of trigeminal nerve.



Mandibular Nerve

- **Largest division** of **trigeminal nerve**
- **Mixed nerve** → has both **sensory & motor roots**
- **Sensory root** arises from **trigeminal ganglion**
- **Motor root** lies **deep to trigeminal ganglion**

Course

- Both **roots pass through foramen ovale** → enter **infratemporal fossa**
- Below **foramen ovale**, both roots **unite to form mandibular nerve trunk**
- **Trunk lies** → **medially to tensor veli palatini** & **laterally to lateral pterygoid**
- **Otic ganglion** present on **medial side** of nerve trunk
- **Divides into two divisions**
    - **Anterior division** (mainly motor)
    - **Posterior division** (mainly sensory)

Branches

From Main Trunk

- **Meningeal branch (Nervous spinosus)** → enters **foramen spinosum**, supplies **dura mater**
- **Nerve to medial pterygoid** → supplies **medial pterygoid**, **tensor tympani**, **tensor veli palatini**

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From Anterior Division (Mainly Motor)

- **Nerve to masseter** → supplies **masseter muscle** & **TMJ**
- **Nerve to lateral pterygoid** → supplies **lateral pterygoid muscle**
- **Deep temporal nerves** → supply **temporalis muscle**
- **Buccal nerve** (**Sensory**) → supplies **cheek mucosa & skin**

From Posterior Division (Mainly Sensory)

- **Auriculotemporal nerve**
  - **Auricular branches** → external ear
  - **Temporal branches** → skin of temporal region
  - **Glandular branches** → parotid gland
  - **TMJ branch**
- **Inferior alveolar nerve (Mixed Nerve)**
  - **Enters mandibular foramen** → supplies **pulp & periodontium of mandibular teeth**
  - **Nerve to mylohyoid (Motor)** → supplies **mylohyoid & anterior belly of digastric**
  - **Branches to molars & premolars** → forms **inferior dental plexus**
  - **Incisive branch** → supplies **incisors & canine teeth**
  - **Mental nerve** → emerges from **mental foramen**, supplies **lower lip & chin**
- **Lingual nerve (Sensory)**
  - **Joined by chorda tympani** of **facial nerve**
  - **Lies medial to mandibular third molar**
  - **Crosses hyoglossus muscle**
  - **Loops around submandibular duct** → supplies **anterior 2/3rd of tongue, floor of mouth, mandibular gums**

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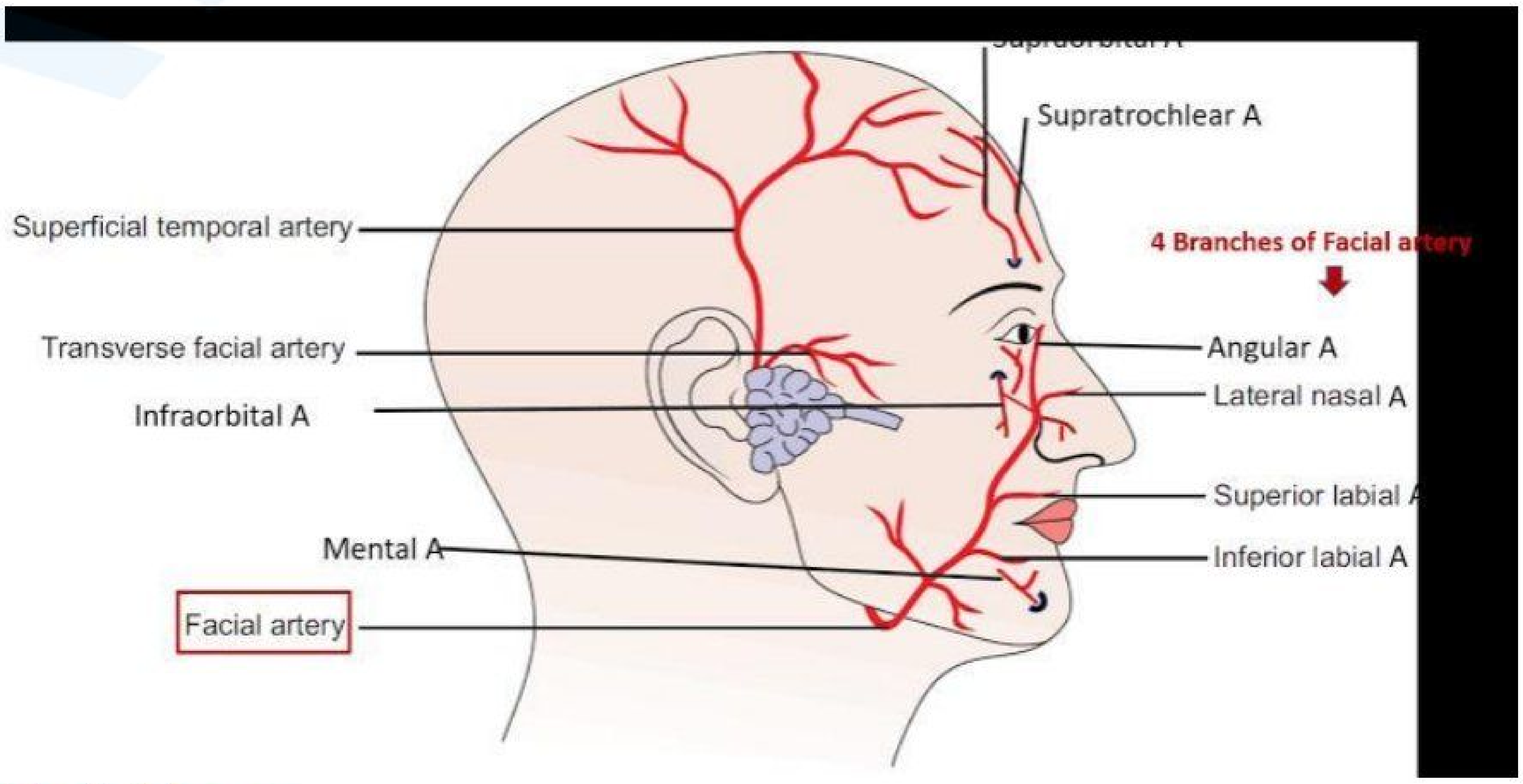
Q5. Describe the origin, course, relation and branches of facial artery.



Facial Artery

- **Main artery of the face** → located **within facial expression muscles**
- **Tortuous course** in the face

Origin

- Arises from **external carotid artery**
- **Above greater horn of hyoid bone** in **carotid triangle**

Course

- **Passes upwards**, grooves **submandibular gland**
- Turns **downwards lateral to submandibular gland**
- **Pierces stylomandibular ligament**
- **Winds around base of mandible** → pierces **deep cervical fascia** at **anteroinferior angle of masseter** → called **'Anaesthetist's artery'**
- Runs **upwards & forwards** → **1.25 cm lateral to angle of mouth**
- **Ascends beside nose** → reaches **medial angle of eye**
- **Terminates** by supplying **lacrimal sac** & anastomosing with **dorsal nasal branch of ophthalmic artery**

Branches in Neck

- **Ascending palatine artery**
- **Tonsillar artery** → supplies **palatine tonsil**
- **Glandular branches** → supply **submandibular gland & lymph nodes**

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- **Submental artery**

Branches on Face

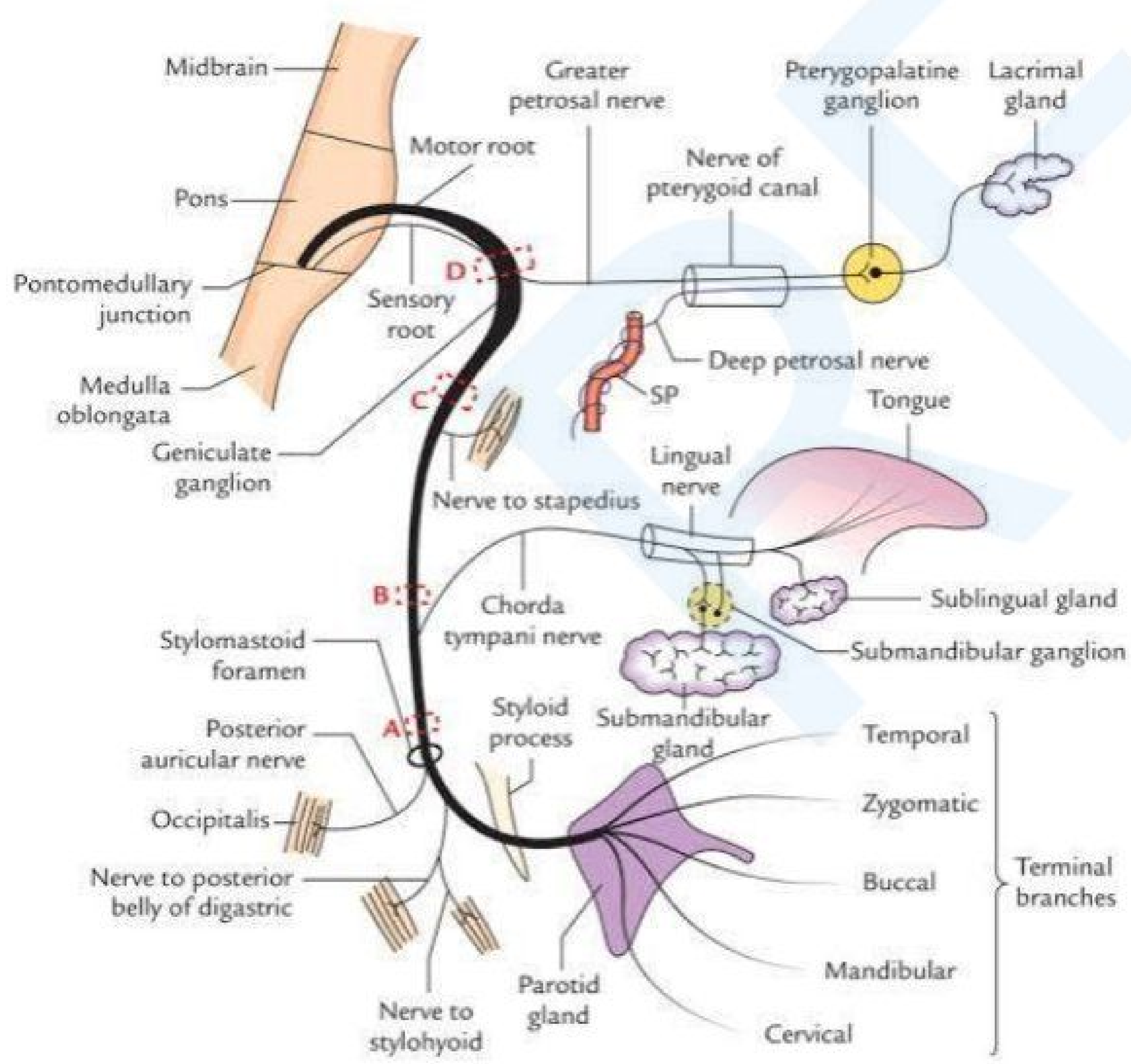
- **Inferior labial artery** → supplies **lower lip**
- **Superior labial artery** → supplies **upper lip & nose**
- **Lateral nasal artery**
- **Muscular branches** → supply **facial expression muscles**

Applied Anatomy

- **Facial artery compression** against **lower border of mandible** can **stop bleeding**
- **Labial branches** are **closer to mucous membrane than skin**
- **Tooth injuries** can cause **hidden hematomas in lips**
- **Labial arteries** on **both sides anastomose** → **connect external carotid arteries of both sides**

Q6. Give an account on Course , intracranial and extra cranial course and applied anatomy of Facial nerve

Course of Facial Nerve



- **Origin**
  - Arises from the **pons** at the **pontomedullary junction**.
  - Consists of two roots:

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        - **Motor root** → Supplies muscles of facial expression.
        - **Nervus intermedius** → Carries taste and secretomotor fibers.
    - Both roots travel together into the **internal acoustic meatus** in the **petrous part of the temporal bone**.
- **Intracranial Course**
    - Enters the **facial canal** within the temporal bone.
    - Forms the **geniculate ganglion** at the first bend in the facial canal.
    - Three branches arise:
        - **Greater petrosal nerve** → Supplies **lacrimal gland** and **nasal mucosa**.
        - **Nerve to stapedius** → Supplies **stapedius muscle** (controls sound vibrations).
        - **Chorda tympani** → Joins the **lingual nerve** to provide taste sensation from the **anterior two-thirds of the tongue** and secretomotor fibers to the **submandibular and sublingual glands**.
    - The nerve bends downward and exits the skull via the **stylomastoid foramen**.
- **Extracranial Course**
    - Emerges from the **stylomastoid foramen** and gives the following branches:
        - **Posterior auricular nerve** → Supplies muscles of the external ear and occipitalis.
        - **Branches to digastric and stylohyoid muscles**.
    - Enters the **parotid gland** and divides into five terminal branches:
        - **Temporal branch** → Supplies muscles of the forehead.
        - **Zygomatic branch** → Supplies **orbicularis oculi** (closes the eye).
        - **Buccal branch** → Supplies **upper lip** and **buccinator muscle**.
        - **Marginal mandibular branch** → Supplies **lower lip muscles**.
        - **Cervical branch** → Supplies **platysma muscle**.
- **Functions of the Facial Nerve**
    - **Motor** → Muscles of facial expression.
    - **Secretomotor** → Lacrimal, submandibular, and sublingual glands.
    - **Taste** → Anterior two-thirds of the tongue.
    - **Sensory** → External ear.
- **Applied Anatomy**
    - **Bell's Palsy** → Sudden paralysis of facial muscles due to nerve damage.
    - **Crocodile Tears Syndrome** → Miswiring of secretomotor fibers, causing tearing while eating.
    - **Ramsay Hunt Syndrome** → Viral infection affecting the geniculate ganglion, leading to facial paralysis and ear pain.

Q7. Give a short account of parts, relation and nerve supply of lacrimal gland

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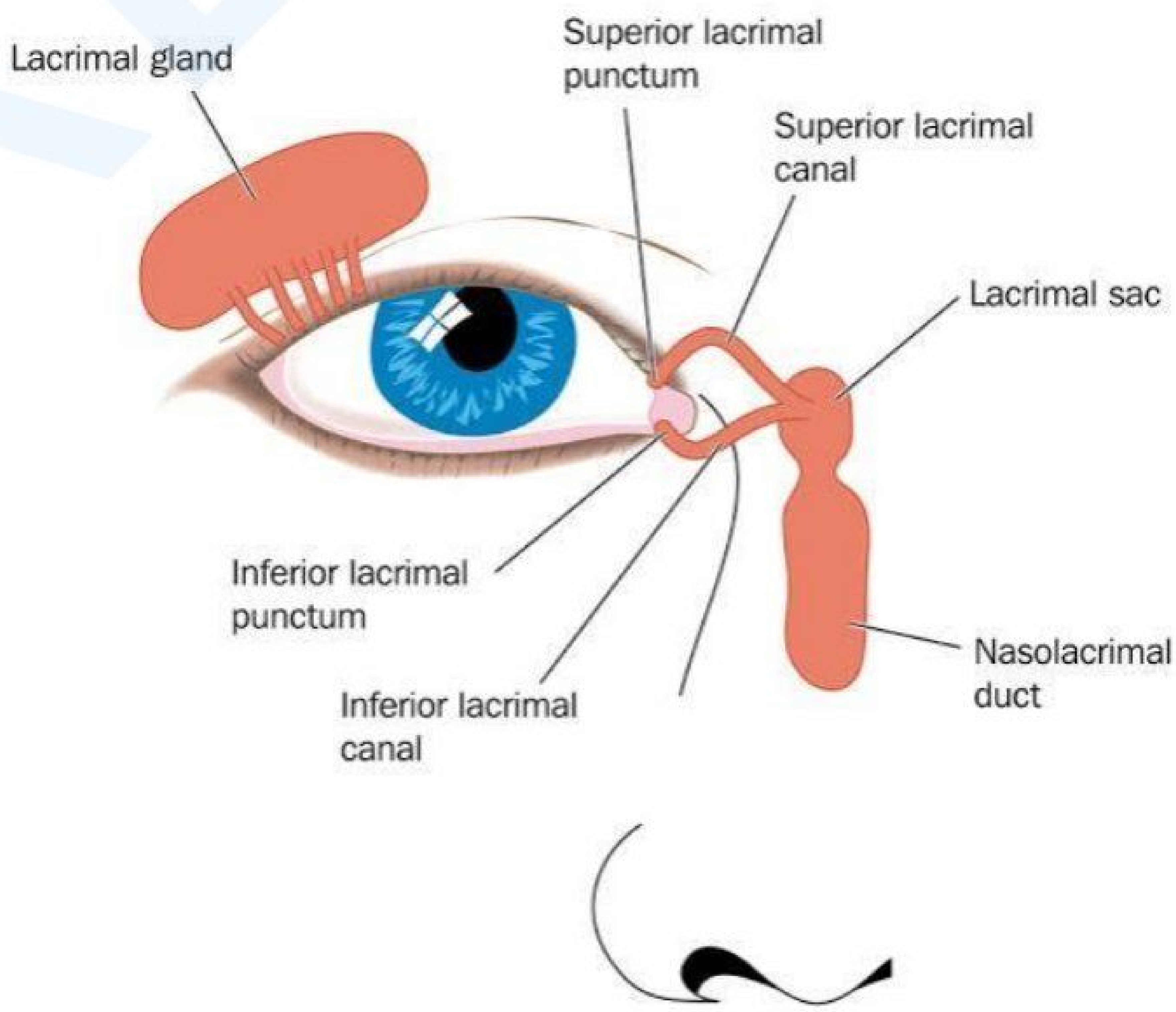

Lacrimal Gland

Lacrimal Gland:

- A **serous gland** that is **'J' shaped**.
- Indented by the tendon of the **levator palpebrae superioris muscle**.
- Located mainly in the **lacrimal fossa** on the lateral part of the **roof of the bony orbit** and partly on the **upper eyelid**.
- **Small accessory lacrimal glands** are found in the **conjunctival fornices**.

Parts of the Lacrimal Gland:

- **Palpebral Part**:
  - Smaller and superficial.
  - Lies within the **eyelid**.
- **Orbital Part**:
  - Larger and deeper.

Ducts:

- About a dozen ducts pierce the **conjunctiva** of the upper lid.
- Open into the **conjunctival sac** near the **superior fornix**.
- Most ducts of the **orbital part** pass through the **palpebral part**.

Blood Supply and Innervation:

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- Supplied by the **lacrimal branch** of the **ophthalmic artery**.
- Innervated by the **lacrimal nerve** (has both **sensory** and **secretomotor fibers**).
- **Course of Secretomotor Fibers**:

Lacrimatory Nucleus
↓
Nerves Intermedius
↓
Geniculate Ganglion
↓
Greater Petrosal Nerve
↓
Nerve of Pterygoid Canal
↓
Pterygopalatine Ganglion
↓
Relay
↓
Zygomatic and Zygomaticotemporal Nerve
↓
Lacrimal Nerve
↓
Lacrimal Gland

Lacrimal Fluid:

- Secreted by the **lacrimal gland**.
- Flows into the **conjunctival sac**, lubricating the **front of the eye** and the **deep surface of the lids**.
- Most fluid **evaporates**; the rest is drained by **lacrimal canaliculi**.
- When excessive, it **overflows as tears**.

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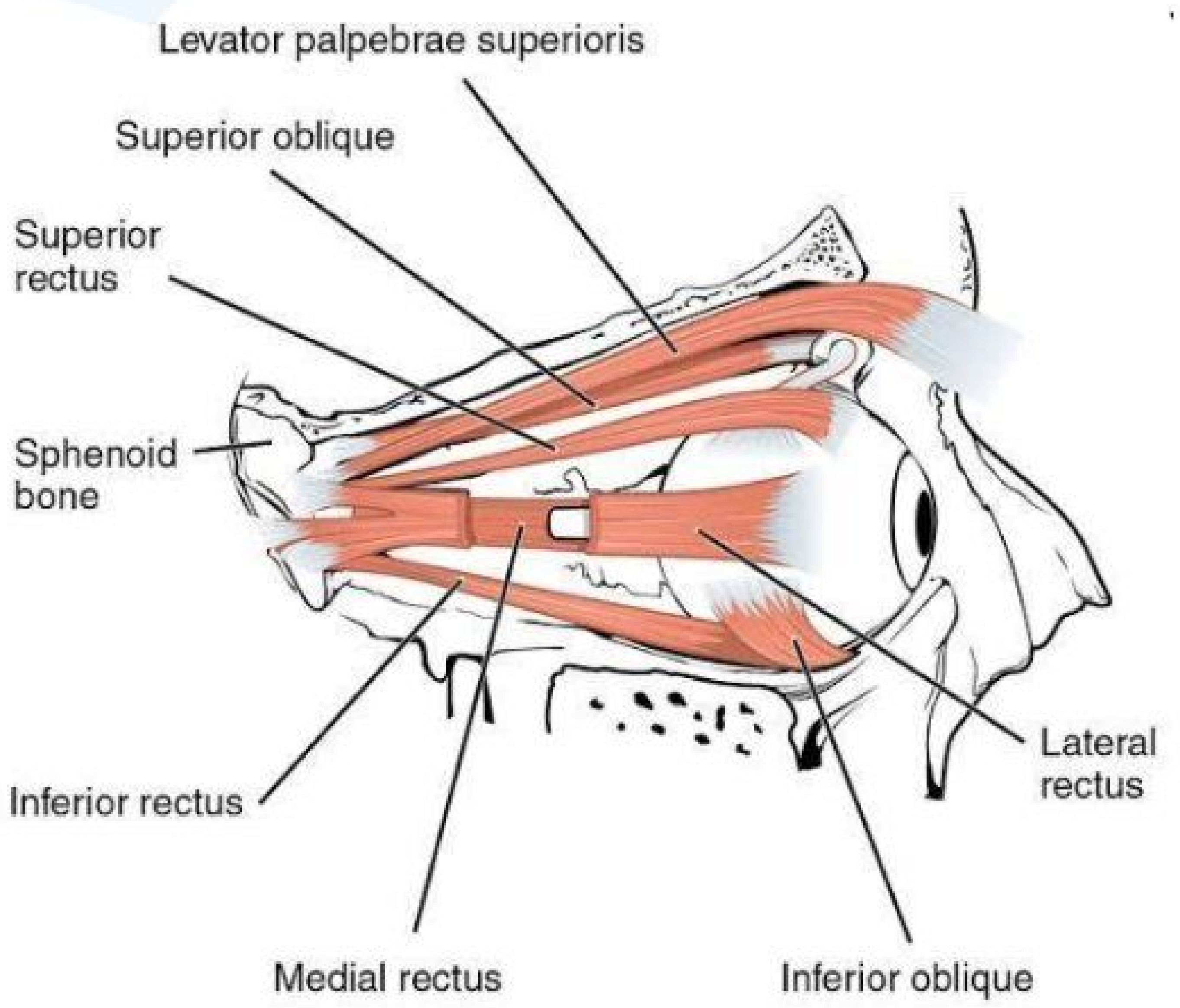
Q8. Levator Palpebrae Superioris Muscle



• **Eye opener muscle (Dilator muscle)**

Origin

1. **Inferior surface** of the **lesser wing** of the **sphenoid bone**
2. **Orbital surface** of the **body** of the **sphenoid bone**, anterior to the **optic foramen**

Insertion

1. Expands towards the **upper eyelid** and divides into **superficial** and **deep layers**, inserted into:
    - **Orbital septum** and **palpebral ligaments** (medial & lateral)
    - **Superior tarsal plate**
    - **Skin of the upper eyelid**
2. Smooth muscle fibres of the **levator palpebrae superioris** are called **Müller's muscle**
3. The **aponeurosis** of this muscle divides the **lacrimal gland** into **palpebral** and **orbital parts**

Nerve Supply

1. **Upper division** of the **oculomotor nerve**
2. **Sympathetic fibres** from **T1 segment** of the **spinal cord** supply the **smooth muscle fibres**

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Actions

• **Elevates** the **upper eyelid**

Applied Anatomy

1. **Paralysis** of the muscle causes **ptosis** (**drooping** of the **upper eyelid**)
2. **Paralysis** of the **smooth muscle fibres** causes **pseudoptosis** (**Horner's syndrome**)

Some Important Terms:

Platysma

1. **Platysma** is one of the **muscles of the neck.**
2. Originates from the **upper parts of pectoral and deltoid fasciae**; fibers run **upwards and medially.**
3. Anterior fibers insert into the **base of the mandible**; posterior fibers insert into the **skin of the lower face and lip**, may be continuous with the **risorius**.
4. **Actions**:
   - Releases pressure of skin on the **subjacent veins.**
   - **Depresses the mandible**.
   - Pulls the angle of the mouth **downwards** as in **horror** or **surprise**.

Buccinator

1. **Buccinator** is the muscle of the **cheek**.
2. **Origin**:
   - Upper fibers: from **maxilla** opposite **molar teeth**.
   - Lower fibers: from **mandible**, opposite **molar teeth.**
   - Middle fibers: from **pterygomandibular raphe**.
3. **Insertion**:
   - Upper fibers: straight to the **upper lip**.
   - Lower fibers: straight to the **lower lip**.
   - Middle fibers decussate before passing to the **lips**.
4. **Actions**:
   - Flattens **cheek** against **gums and teeth**.
   - Prevents accumulation of food in the **vestibule**.

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Facial Vein

1. The **facial vein** is the **largest vein** of the face with **no valves**.
2. Begins as the **angular vein** at the **medial angle** of the eye.
3. Formed by the union of the **supratrochlear** and **supraorbital veins**; continues as the **facial vein**, running downwards and backwards behind the **facial artery**.
4. Crosses the **anteroinferior angle** of the **masseter**, pierces the **deep fascia**, crosses the **submandibular gland**, and joins the **anterior division** of the **retromandibular vein** below the angle of the mandible to form the **common facial vein**, which drains into the **internal jugular vein**.

Emissary Vein

- **Definition**: Veins connecting **intracranial venous sinuses** to **extracranial veins**.
- **Function**: Aids in **venous drainage** from **skull** to **scalp**.
- **Types**:
    - **Occipital**: Connects **occipital sinus** with **neck veins**.
    - **Parietal**: Connects **superior sagittal sinus** with **scalp veins**.
    - **Mastoid**: Connects **sigmoid sinus** with **ear veins**.
- **Clinical Relevance**: Can spread **infections** between **cranial cavity** and **external environment**.

Orbicularis Oculi

1. **Orbicularis oculi** is composed of three parts:
    - **Orbital Part**:
        - Origin: Part of **medial palpebral ligament** and adjoining **bone**.
        - Insertion: Concentric rings return to the point of origin.
        - **Action**: Closes lids **tightly**; protects eye from **bright light**.
    - **Palpebral Part**:
        - Origin: **Lateral part** of **medial palpebral raphe**.
        - Insertion: **Upper and lower eyelids**.
        - **Action**: Closes lids **gently; blinking**.
    - **Lacrimal Part**:
        - Origin: **Lacrimal fascia** and **lacrimal bone**.
        - Insertion: **Upper and lower eyelids**.
        - **Action**: **Dilates lacrimal sac;** supports the **lower eyelid**.

Palpebral Ligament

1. The **palpebral fascia** of the two lids forms the **orbital septum**.

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2. Its **thickenings** form **tarsal plates** of **tarsi** in the lids and the **palpebral ligaments** at the angles.

Dangerous Area of Scalp

1. Infections from the face can spread in a **retrograde direction** and cause **thrombosis** of the **cavernous sinus**.
2. This is likely in the presence of infection in the **upper lip** and **lower part** of the **nose**. This area is therefore called the **dangerous area of the face**.

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